



MAX PLANCK INSTITUTE
FOR SOFTWARE SYSTEMS



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**UNIVERSITÄT
DES
SAARLANDES**



Hellinger Multimodal Variational Autoencoders

Presenter: Huyen Vo

AISTATS 2026 - Spotlight

Huyen Vo

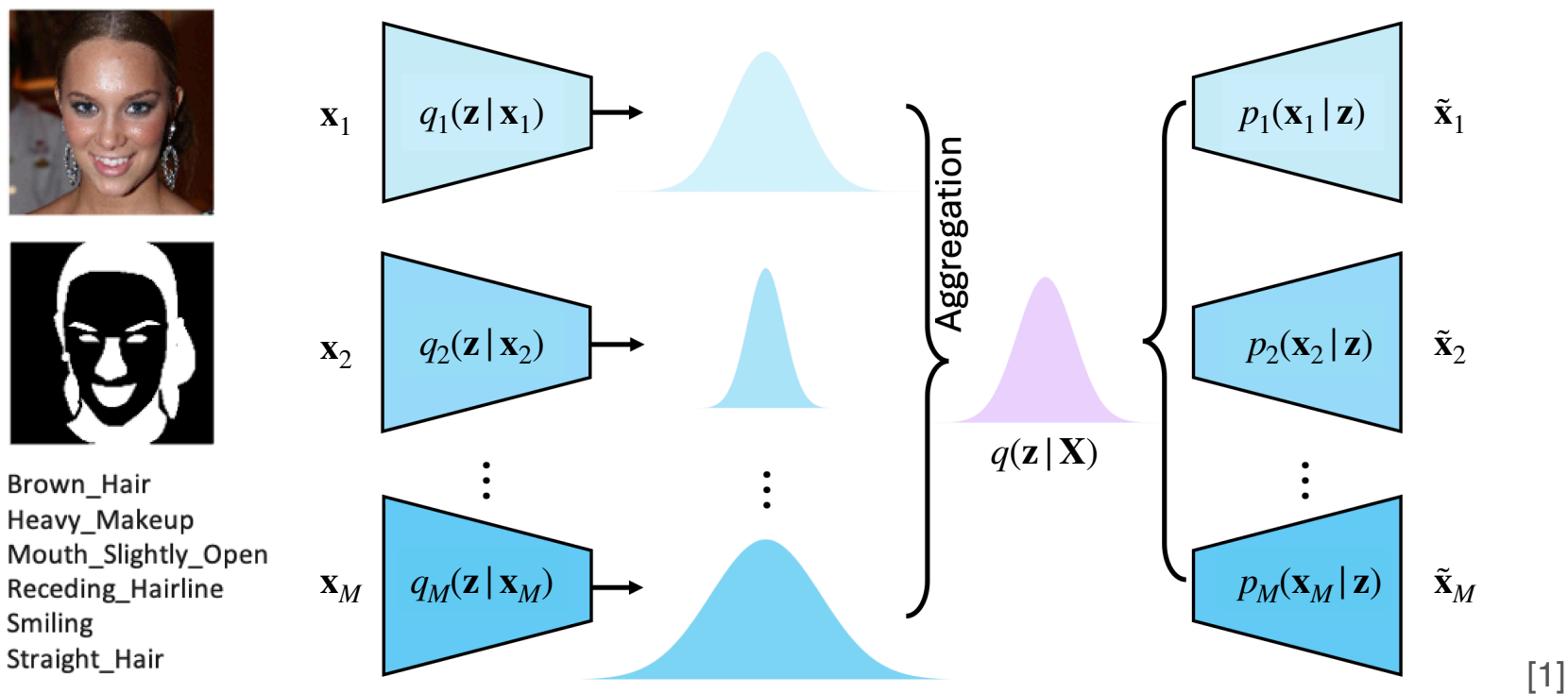


Isabel Valera



Motivation

In multimodal generative learning

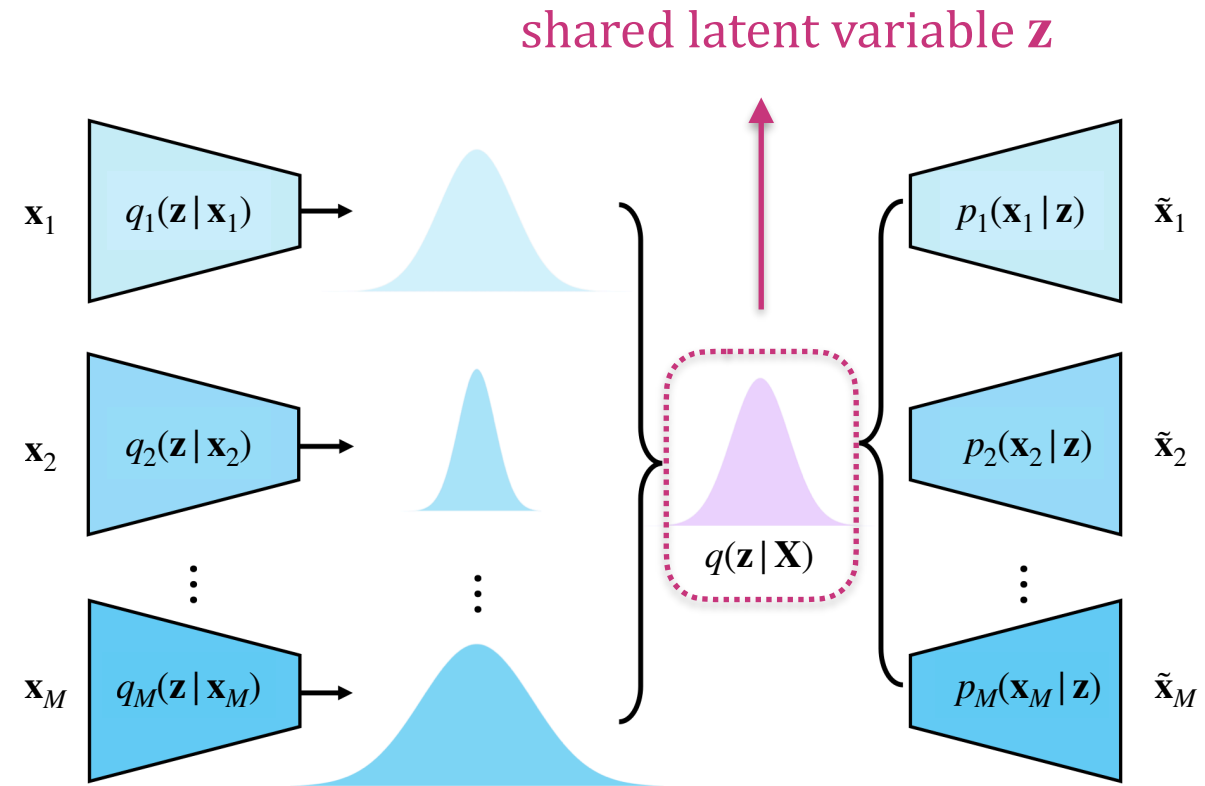


[1] Qiu et al. Multimodal Variational Autoencoder: a Barycentric View.

Motivation

We want **multimodal generative models** that support

1. learn shared representations,



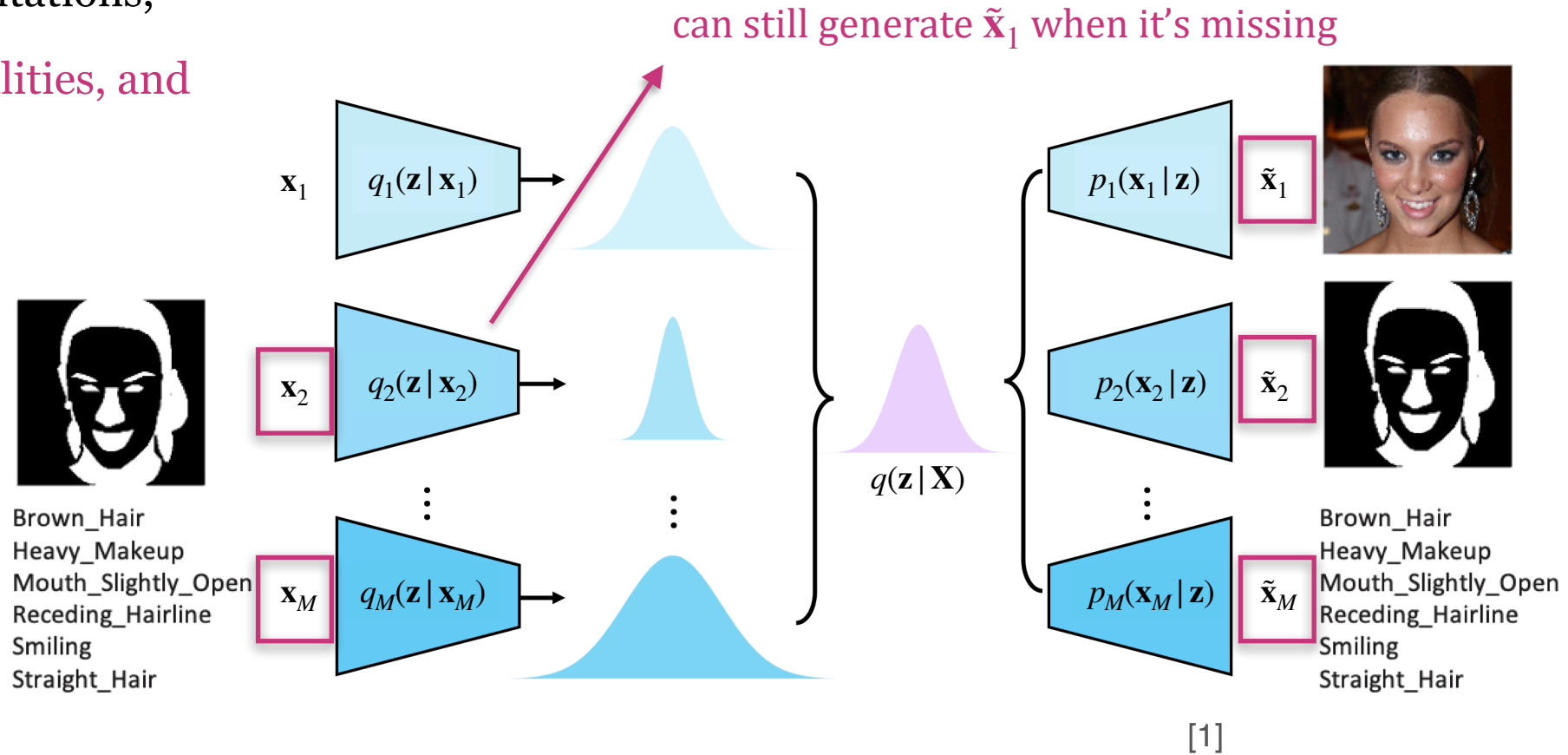
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We want **multimodal generative models** that support

1. learn shared representations,
2. handle missing modalities, and

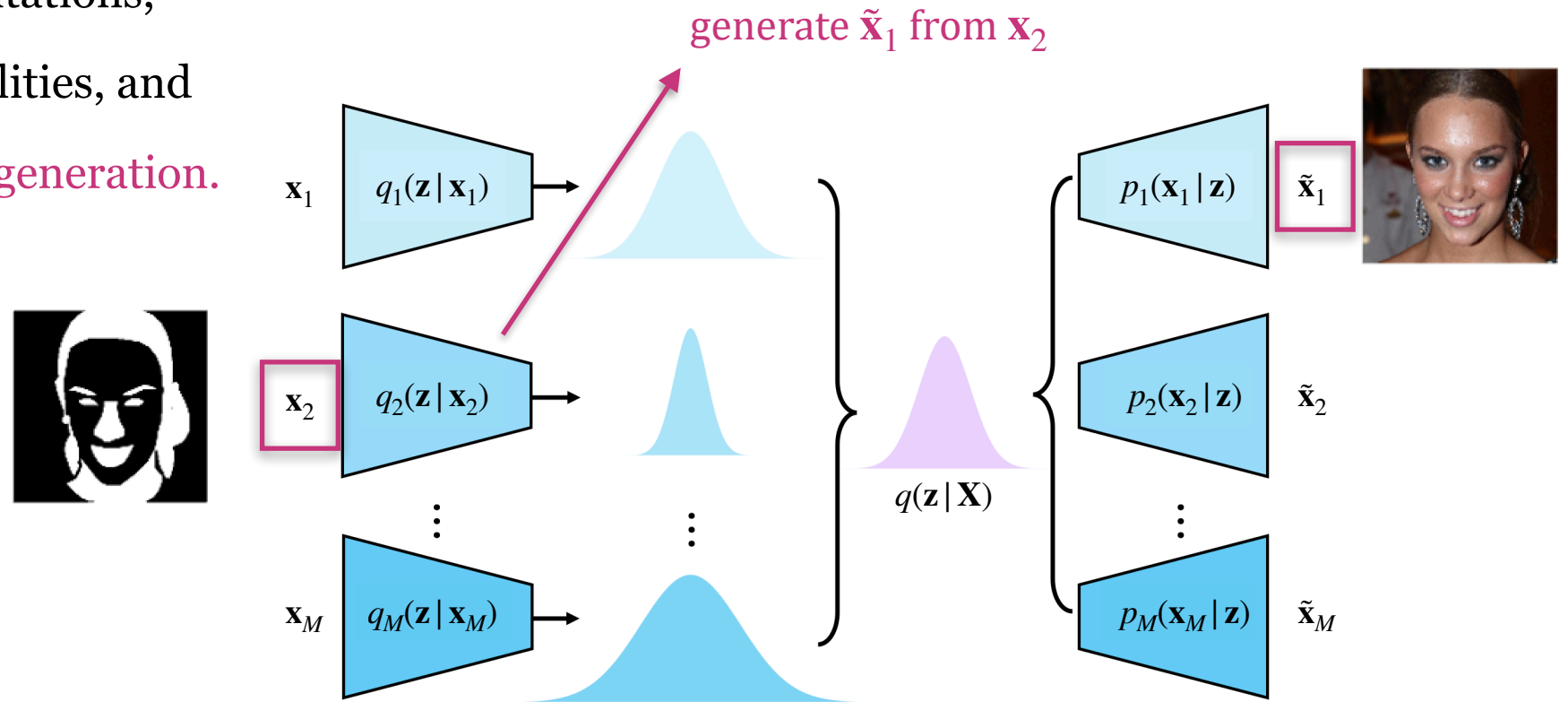


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Motivation

We want **multimodal generative models** that support

1. learn shared representations,
2. handle missing modalities, and
3. support cross-modal generation.

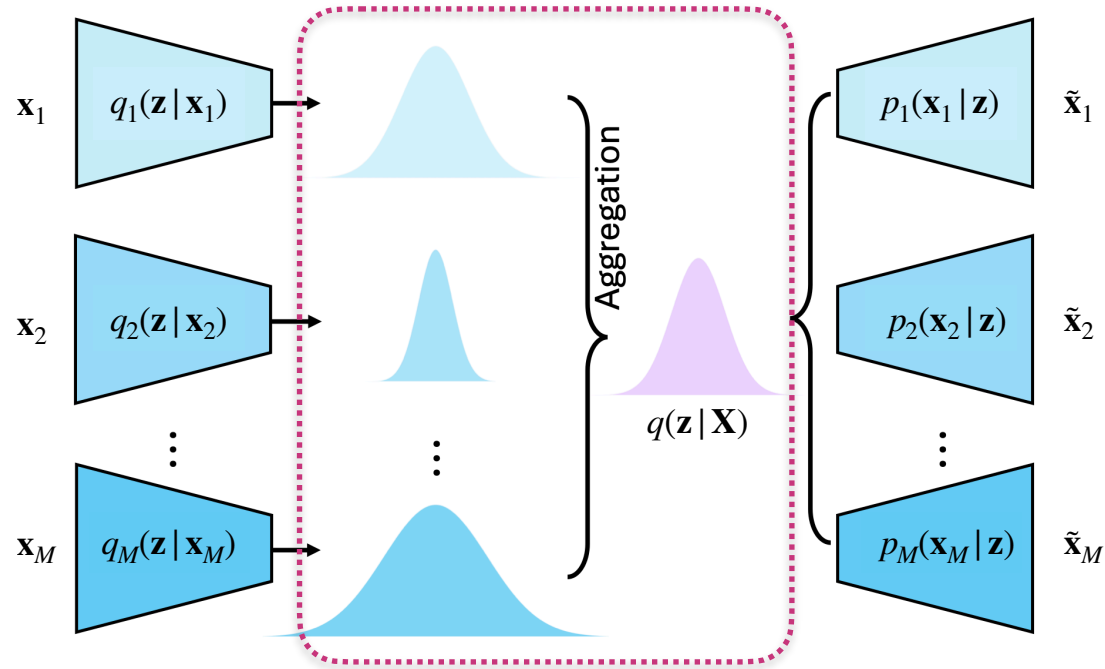


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Motivation

Posteriors as **Gaussian** distributions



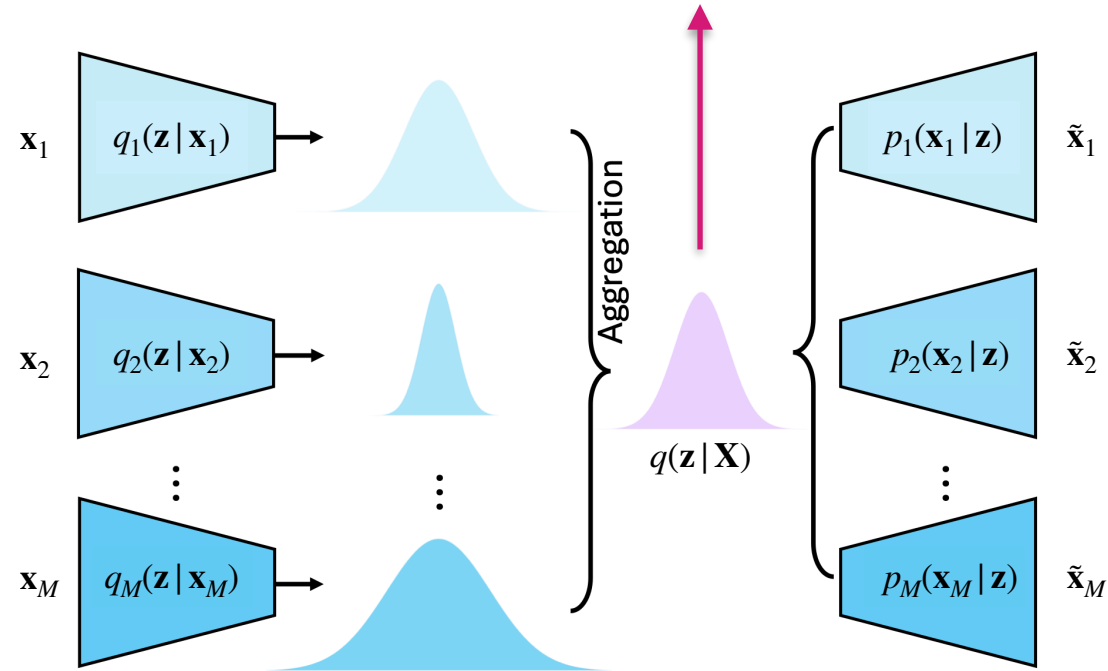
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→ Multimodal VAEs learn **a joint posterior** $q(\mathbf{z} | \mathbf{X})$ from M modalities using
 M encoders $\{q_j(\mathbf{z} | \mathbf{x}_j)\}_{j=1}^M$ and M decoders $\{p_j(\mathbf{x}_j | \mathbf{z})\}_{j=1}^M$

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Motivation

How to approximate the joint posterior $q(\mathbf{z} | \mathbf{X})$?



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Current Gaps

Product of experts (PoE):

Mixture of experts (MoE):

Current Gaps

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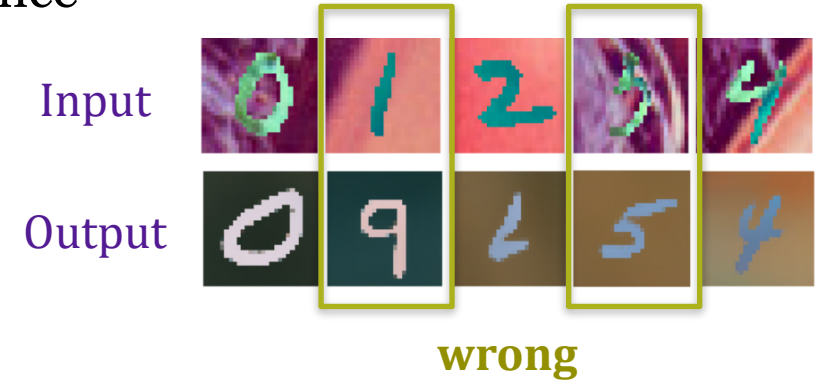
$$q(\mathbf{z} | \mathbf{X}) = c \prod_{j=1}^M q_j(\mathbf{z} | \mathbf{x}_j)$$

Mixture of experts (MoE):

Current Gaps

Product of experts (PoE): ✓ high quality, ✗ low coherence

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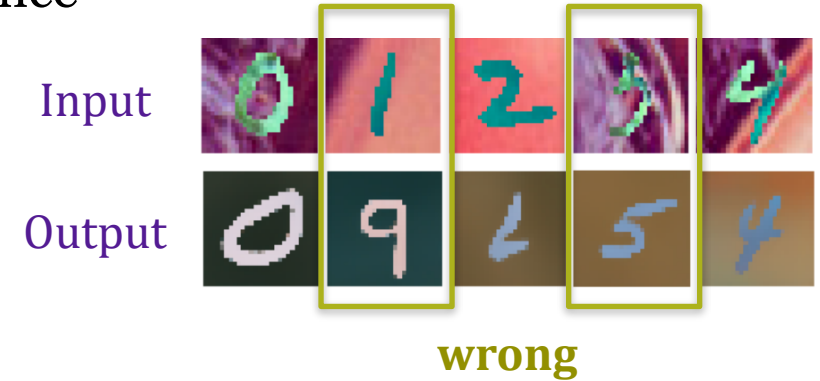


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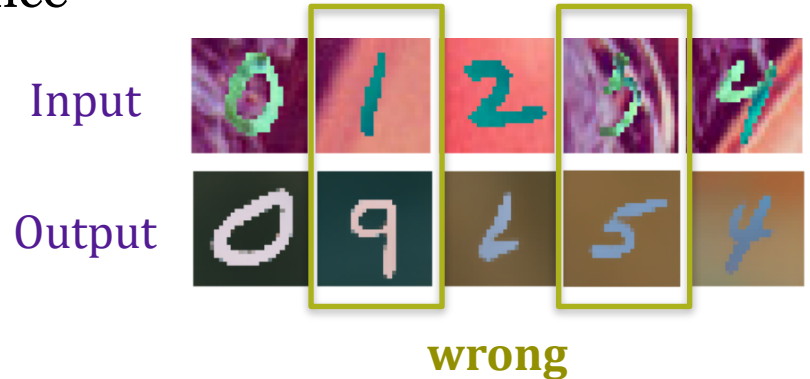
Mixture of experts (MoE):

$$q(\mathbf{z} | \mathbf{X}) = \frac{1}{M} \sum_{j=1}^M q_j(\mathbf{z} | \mathbf{x}_j)$$

Current Gaps

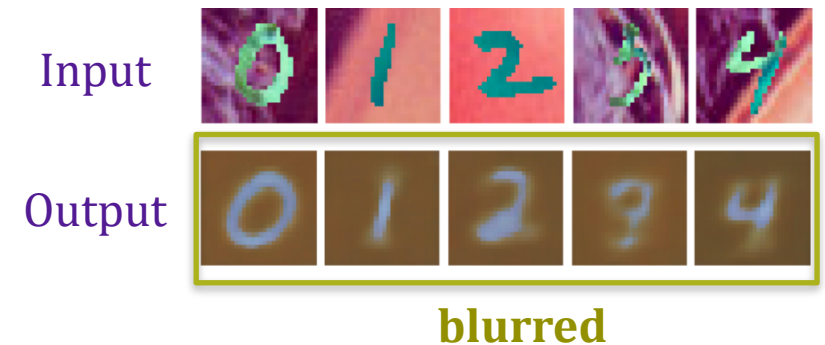
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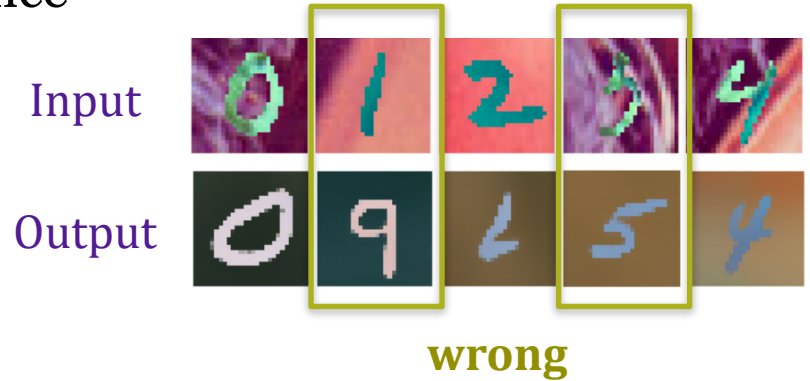
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→ a trade-off between quality and coherence

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Product of experts (PoE): ✓ high quality, ✗ low coherence

Mixture of experts (MoE): ✗ low quality, ✓ high coherence

→ a trade-off between quality and coherence

Research Question

Can we design a model that achieves
both high quality and high coherence while remaining efficient?

Multimodal aggregation as Probabilistic Opinion Pooling

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Aggregating experts by minimizing the discrepancy between the individual and aggregated distributions.

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Using the α -divergence yields the closed-form solution known as **Hölder pooling**:

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$$\alpha \rightarrow 0$$

(reverse KL)

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Both measures are
asymmetric and unbounded

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asymmetric & unbounded

✓ high quality, ✗ low coherence

$$\alpha = 0.5$$

unique symmetric

bounded

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**unique symmetric
bounded**

✓ high quality ✓ high coherence ??

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Mixture of experts (**MoE**)

asymmetric & unbounded

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Hellinger aggregation in Multimodal VAEs

Hölder pooling ($\alpha = 0.5$)

A mixture-of-Gaussians form

× Requires sub-sampling

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Hellinger aggregation

a single-Gaussian approximation of Hölder pooling ($\alpha = 0.5$) via moment matching,

$$\tilde{\mu} = \int \mathbf{z} q(\mathbf{z}) d\mathbf{z}, \quad \tilde{\Sigma} = \int \mathbf{z} \mathbf{z}^\top q(\mathbf{z}) d\mathbf{z} - \tilde{\mu} \tilde{\mu}^\top$$

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✓ No sub-sampling

Hellinger aggregation

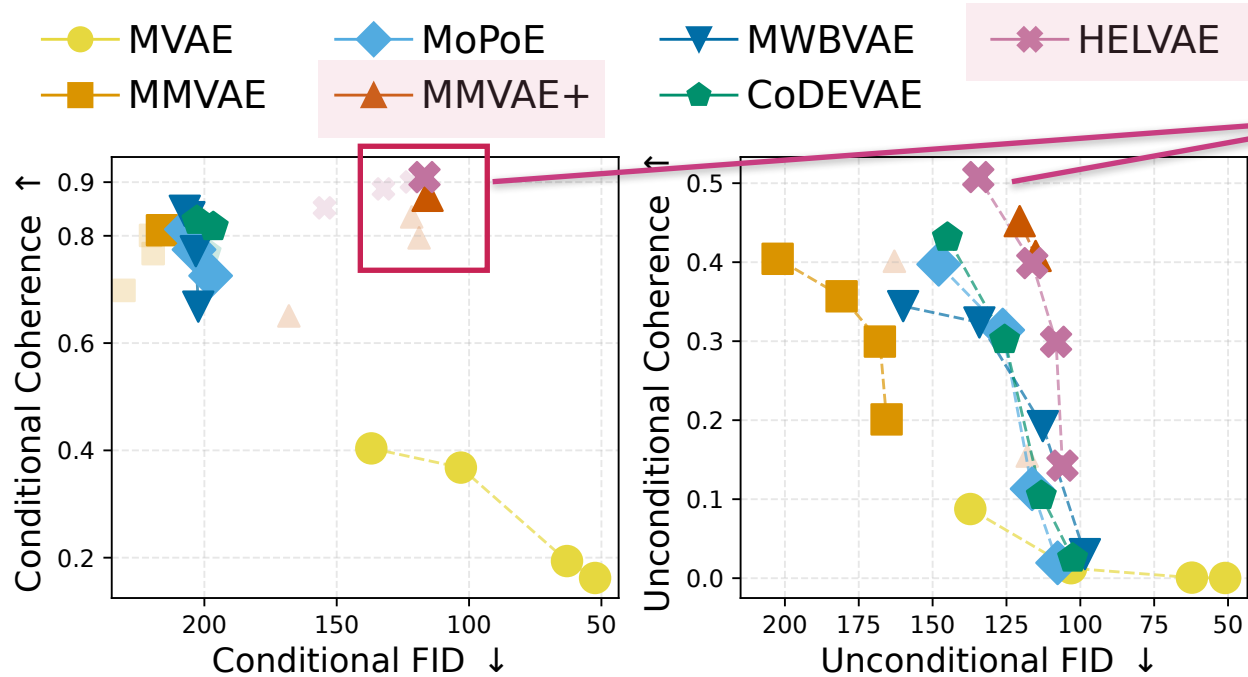
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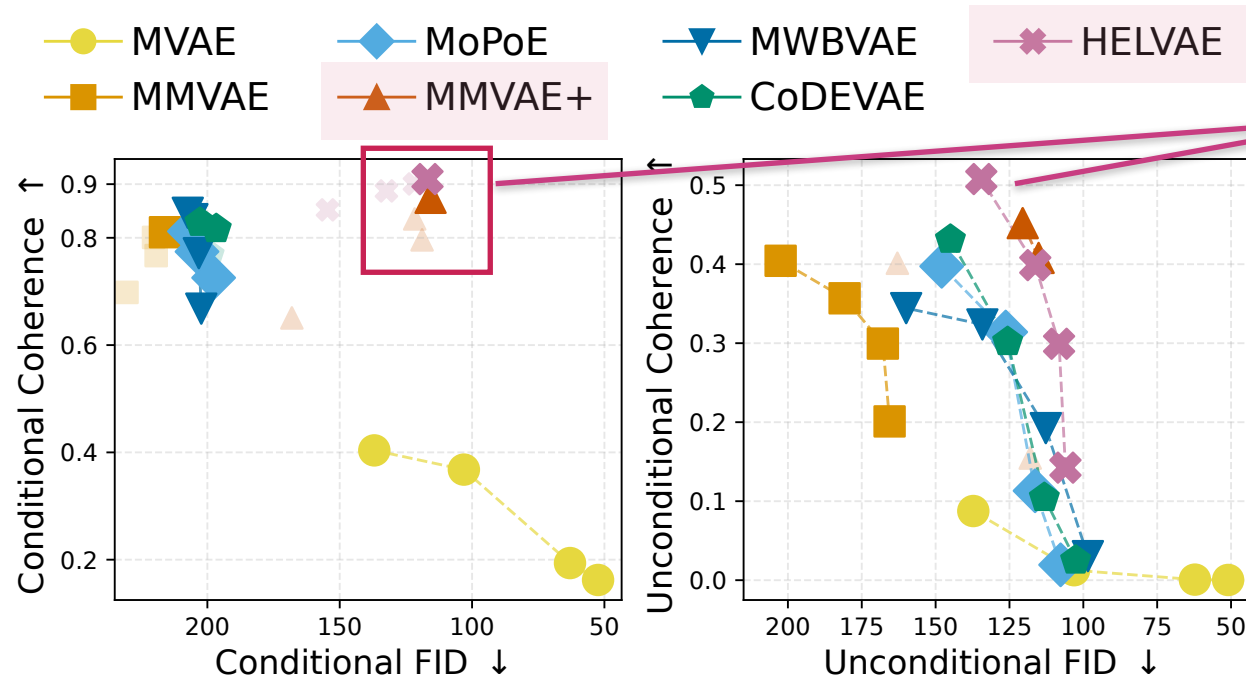
Hellinger aggregation defines a novel multimodal VAE, called **HELVAE**

Experimental Results



HELVAE lies close to the Pareto frontier,
outperforming the SOTA **MMVAE+**

Experimental Results

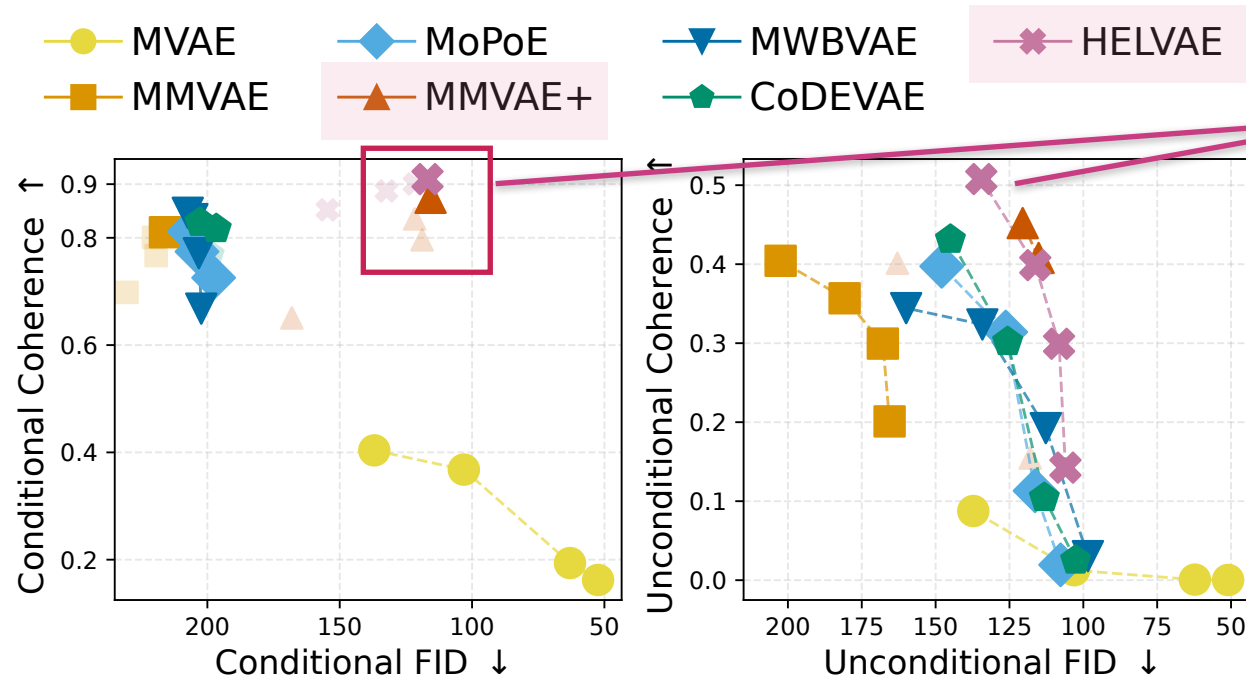


HELVAE lies close to the Pareto frontier, outperforming the SOTA **MMVAE+**

HELVAE	MMVAE+
0.0895 seconds/batch	0.2884 seconds/batch

Up to **3× faster** batch training

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0.0895 seconds/batch	0.2884 seconds/batch

Up to **3× faster** batch training

HELVAE: ✓ high coherence ✓ high quality ✓ computationally efficient
→ an efficient yet effective model



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Paper



Code



UNIVERSITÄT
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Today at **Poster #186**
4pm - 7pm



Hiring Postdoc

Huyen Vo



Isabel Valera

